

AI Agents for Materials Discovery

Retrieval-augmented, property-aware and generative machine learning for inorganic crystal design

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Format note. Drafted to the MSci brief: ≈8 pages of literature review (§1–§7) and ≈2 pages of research proposal (§8–§12), ≤10 pages total. Marking criteria addressed: coverage; insight & context; project background/aims/objectives; programme plan; risk, fall-backs & flexibility.

Abstract

Large language models (LLMs) have made scientific knowledge conversationally accessible, and retrieval-augmented generation (RAG) grounds their answers in trusted sources. Yet LLMs remain unreliable on the *continuous, numerical* reasoning that materials science demands, and pure RAG can *retrieve and summarise* facts but cannot *compute* a property or *propose* a new compound. In parallel, two specialised model families have matured: graph neural networks (GNNs) that predict material properties at a fraction of the cost of density functional theory (DFT), and diffusion-based generative models that produce stable, property-conditioned crystals. This review surveys foundation models, RAG/agent frameworks, property-prediction networks and generative models for inorganic materials, and argues that these strands remain largely *siloed*. The proposed project closes that gap by extending a materials RAG framework into a tool-using *LLM agent* that delegates numerical property prediction to dedicated ML models and crystal generation to a diffusion model, turning a literature-retrieval assistant into an end-to-end, query-driven discovery engine.

1. Introduction and motivation

Materials science is entering a phase of data-driven discovery. For decades, computational materials research has rested on first-principles methods such as density functional theory (DFT), molecular dynamics and finite-element modelling to predict properties and rationalise mechanisms.¹ These methods are physically principled but computationally expensive and, in practice, confined to small, well-characterised systems; a single DFT relaxation of a modest unit cell can consume hours of CPU time, making exhaustive exploration of chemical space infeasible. Machine learning (ML) trained on curated computational and experimental data has begun to supplement these simulations, enabling property prediction and generative design at a tiny fraction of the cost.^{1,19} Materials data is, however, uniquely demanding: it is intrinsically multimodal (structures, text, spectra, images), spans length scales from the electronic to the macroscopic, must respect physical laws such as energy conservation and symmetry, and is comparatively scarce and imbalanced relative to the billion-token corpora that train language models.¹ Historically, too, such models have been narrow: each property, material class or data modality required a bespoke architecture and training pipeline, limiting transfer and scalability.

The arrival of **foundation models** (FMs), large models pretrained on broad data that transfer across many downstream tasks, has reframed the field.³⁵ Following the impact of the Transformer⁴ and its descendants BERT⁵ and GPT⁶ in natural-language processing, the materials community now asks whether similarly general models can unify reasoning over atomic structures, chemical formulae, spectra and the text of the literature.¹ Two capabilities have proved especially consequential. First, **retrieval-augmented generation** (RAG) couples an LLM to an external knowledge store so that responses are grounded in retrieved evidence rather than parametric memory alone, mitigating hallucination.⁷ Second, **LLM agents** wrap a language model in a reasoning loop that can call external tools,

query databases and act on the results, allowing the model to outsource tasks it cannot perform reliably on its own.³⁰

This last point is the crux of the project. LLMs are powerful interfaces to *qualitative* knowledge but are demonstrably weak at the *quantitative*, continuous reasoning at the heart of materials science: predicting a band gap, a formation energy or an elastic modulus to useful accuracy.³¹ A RAG system can find that a compound's band gap was reported as 1.6 eV; it cannot, by itself, predict the band gap of a compound that has never been measured, nor can it generate a novel crystal that should exhibit a target band gap. Those are precisely the tasks at which specialised ML such as GNN property predictors^{19,20} and diffusion-based crystal generators^{12,13} now excels. This review therefore traces a deliberate arc: from the general landscape of AI for materials, through RAG and agents, to property prediction and generative modelling, ending at the integration gap that the project will address.

Scope is defined by the project. The review concentrates on **inorganic crystalline materials** and on the four capabilities the project must combine: knowledge retrieval, agentic orchestration, property prediction, and property-conditioned generation. It does not attempt the comprehensive coverage of any one of these mature subfields; rather, following the assignment's guidance, it foregrounds the key ideas, representative methods and current thinking needed to situate and justify the proposed work. A recent survey of AI for materials science¹ provides the organising backbone and is supplemented throughout with the primary literature.

2. Foundation models and the AI-for-materials task landscape

It is useful to begin with a map of the problems that AI is being applied to in materials science, because the project sits at the intersection of several of them. Van *et al.*¹ organise the field into six recurring task families (Table 1); the project draws principally on T1, T3 and T4, orchestrated by an agent.

Table 1. The six task families that recur across AI-for-materials research, after the taxonomy of Van *et al.*¹ Tasks central to this project are highlighted.

Task	Family	What it involves	Project
T1	Data extraction, interpretation & Q&A	Mining facts and relations from papers/patents; question answering over the literature, the natural home of RAG.	<i>core</i>
T2	Atomistic simulation	Machine-learned interatomic potentials replacing/accelerating DFT and MD.	context
T3	Property prediction	Mapping composition/structure to properties (band gap, formation energy, moduli).	<i>core</i>
T4	Structure, design & discovery	Generating novel, stable, property-targeted structures (inverse design).	<i>core</i>
T5	Process planning & optimisation	Synthesis route planning, experimental optimisation.	extension
T6	Multiscale modelling	Bridging electronic, atomistic and continuum scales.	out of scope

Cutting across these tasks, FMs are conventionally grouped into three categories.¹ **Unimodal** models operate within a single modality: text (e.g. the BERT-derived MatSciBERT²³) or structure (GNNs over crystal graphs). **Multimodal** models attempt to fuse two or more modalities, for example aligning text with molecular structure, but remain comparatively immature: most deployed models still reason in a single modality.¹ **LLM agents** extend a base model with external knowledge and tools, and are the category the project lives in. Architecturally, the three Transformer variants recur throughout: *encoder-only* models (BERT-style) build representations well-suited to property prediction; *decoder-only* models (GPT-style) generate sequences and underpin string-based crystal generation; and *encoder-decoder* models support translation-like tasks.^{4,5}

The strategic reading of this landscape is that *no single model does everything well*. Text models read the literature but cannot relax a structure; potentials and GNNs compute properties but cannot converse; generators propose structures but neither explain their reasoning nor guarantee stability. This division of labour is exactly why an *agentic* architecture, a language model that routes sub-problems to specialised tools, is the natural unifying abstraction, and why it forms the spine of the proposed system.

3. Retrieval-augmented generation and LLM agents for materials

3.1 Why retrieval, and why grounding matters

LLMs encode a great deal of materials knowledge in their parameters, but that knowledge is static, opaque and prone to *hallucination* when the model is pushed outside its training distribution.¹ RAG addresses this by retrieving relevant documents at query time and conditioning generation on them, so that claims are traceable to sources.⁷ In a domain where an incorrect synthesis condition can be unsafe and an invented property value is worse than no answer, this grounding is not a convenience but a requirement. The benchmark study of Miret and Krishnan³¹ is sobering here: general-purpose LLMs are *not yet* reliable for real-world materials discovery, particularly on quantitative tasks. Dedicated benchmarks sharpen the point: when asked to regress material properties directly, LLMs trail task-specific models by a wide margin,^{43,38} evidencing the project's central premise that numerical work should be *delegated* to specialised ML rather than learned in context, while retrieval and tool-use, not ever-larger parametric models, are the productive path.

3.2 RAG systems for materials knowledge

The clearest exemplar is **HoneyComb**,²⁴ an LLM agent built around three components: *MatSciKB*, a curated knowledge base aggregating arXiv articles, Wikipedia, datasets and textbooks; *ToolHub*, a suite of retrieval tools for up-to-date information; and a *Retriever* that pulls relevant knowledge using classical and neural retrieval (BM25 and Contriever). HoneyComb is model-agnostic and measurably improves domain reasoning over a bare LLM, a template for the retrieval layer of the proposed system. Its predecessor HoneyBee²⁵ showed that instruction-tuning an open LLM on materials text (with one model generating instructions and another verifying them) already lifts domain competence, establishing the data-curation pattern the project can reuse.

3.3 From retrieval to action: tool-using agents

The decisive step beyond retrieval is *tool use*. **ChemCrow**³⁰ equips an LLM with eighteen expert chemistry tools and a chain-of-thought loop (plan, select a tool, act, analyse; the reason-then-act pattern formalised as ReAct³⁹), implemented in the LangChain framework, and thereby performs synthesis planning and other tasks no bare LLM can. Most directly relevant is **LLMatDesign**,²⁶ which is almost a blueprint for this project: it takes a target composition and property in natural language, proposes a modification (substitute, add or remove an element), calls dedicated ML tools (MatDeepLearn, TorchMD-Net³²) to *predict the resulting property*, evaluates the outcome against the target, and iterates in a self-reflective loop, translating instructions into Materials Project API calls along the way. **ChatMOF**²⁷ applies the same agent–toolkit–evaluator pattern to metal–organic frameworks, spanning retrieval, prediction *and* generation. Human-in-the-loop systems such as **MatAgent**²⁸ and **MatPilot**²⁹ add expert oversight and, in MatPilot's case, a path toward robotic execution.

Two observations frame the project. First, these agents already demonstrate the value of delegating numerical work to specialised models, but each hard-wires a particular toolset and task. Second, retrieval and generation are rarely unified in one system: HoneyComb retrieves but does not generate crystals; the diffusion generators of §5 generate but are not wrapped in a conversational, knowledge-grounded interface. The project's contribution is to combine grounded retrieval, numerical prediction and generative design behind a single agent.

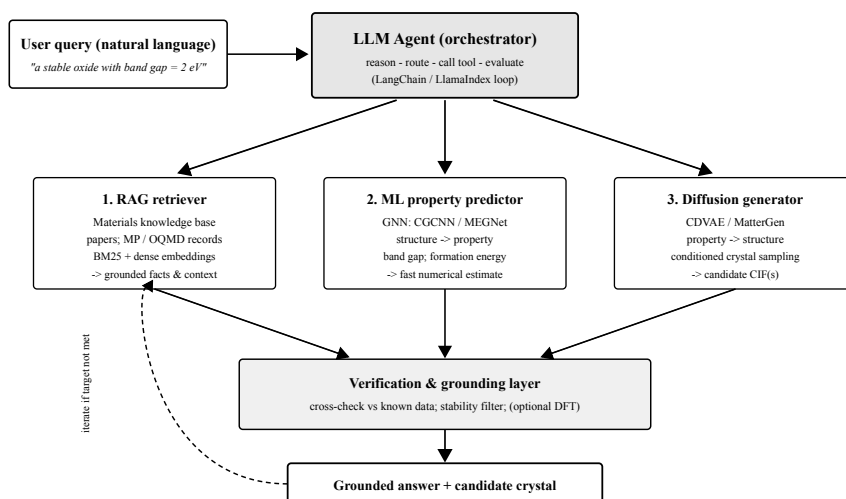


Figure 1. Proposed agentic architecture (original schematic). A single LLM agent interprets a natural-language query and routes sub-tasks to three specialised tools: ① a RAG retriever for grounded knowledge, ② an ML property predictor for fast numerical estimates, and ③ a diffusion model for property-conditioned crystal generation, then verifies candidates against known data before answering, iterating when the target property is unmet.

4. Machine learning for material property prediction

Property prediction is the quantitative engine the agent will call, and the subfield where ML most clearly outperforms an LLM. The premise is simple: DFT is accurate but expensive, so learn the structure→property mapping directly from data and predict in milliseconds.¹ Two model families dominate, distinguished mainly by how they represent a material.^{19,20}

4.1 Graph neural networks over structures

Crystals and molecules are naturally graphs (atoms as nodes, bonds or proximity as edges), and *geometric* GNNs exploit this directly. **SchNet**²¹ introduced continuous-filter convolutions to model quantum interactions with rotational invariance. **CGCNN** (crystal graph CNN)¹⁹ represents a periodic crystal as a multi-edge graph, capturing local chemical environments and long-range periodicity, and is notable for being *interpretable*: it attributes global properties to local environments. **MEGNet**²⁰ generalises this with atomic, bond and global-state attributes and, trained on Materials Project² and QM9 data, predicts formation energies, band gaps and elastic moduli across broad chemical space. These models (fast, accurate and pretrained) are ideal "tools" for an agent, and CGCNN/MEGNet are realistic choices for the project's predictor. A closely related family, machine-learned interatomic potentials such as MACE,¹¹ MatterSim¹⁰ and CHGNet,⁴¹ approximates DFT energies and forces directly and provides the fast energy-above-hull estimates needed to judge thermodynamic stability, a capability the project reuses in its verification stage (§9).

4.2 Transformer- and string-based predictors

An alternative encodes a material as a string and applies a Transformer. **ChemBERTa**²² adapts RoBERTa to molecular SMILES from PubChem for property prediction; related encoder-only models (CatBERTa, SolvBERT, Mol-BERT) target catalysis and solvation. Because SMILES and SELFIES are 2D and omit 3D conformation, crystal-aware variants such as Matformer and ComFormer inject lattice geometry and periodicity into the attention mechanism.¹ The representation question (2D strings vs 3D graphs) is a recurring theme: 3D structural information yields richer features for spatially dependent properties, which is why GNNs remain the default for inorganic crystals.

Crucially for the project, property predictors are also what make *generation* useful: a generator proposes a candidate, and a predictor (or DFT) scores whether it actually meets the target. Standardised benchmarks such as MoleculeNet³³ and Matbench³⁷ allow these predictors to be compared and selected on like-for-like tasks, which the project will use to choose and validate its predictor.

5. Generative and diffusion models for materials

If property prediction answers "what is the property of this material?", generative modelling answers the inverse and far harder question: "what material has this property?" This *inverse design* is the project's most ambitious component. Early generative work used VAEs and GANs; the field has since converged on **diffusion models**, which learn to reverse a gradual noising process to sample new structures⁸ and have proved especially well-suited to the geometric, periodic nature of crystals. In the crystal setting the denoising network is made *equivariant* and operates jointly on the lattice, fractional coordinates and atom types, while property conditioning (e.g. classifier-free guidance) steers sampling toward a target value, the mechanism that makes "generate a crystal with property Y" tractable.^{14,12}

5.1 Graph- and diffusion-based crystal generation

G-SchNet¹⁸ pioneered autoregressive 3D molecule generation with built-in rotational invariance. **CDVAE** (crystal diffusion VAE)¹³ was a landmark for inorganic solids, combining a periodic GNN encoder, a property predictor and a periodic GNN decoder with a diffusion process to generate *stable* crystals, and demonstrating reconstruction, generation and, critically, *property optimisation*. Its conditional extension **Con-CDVAE**¹⁶ generates latent variables from a target property (e.g. formation energy or band gap) and decodes them to a structure, realising the exact "generate a crystal with this property" capability the project envisages. **DiffCSP**¹⁴ performs joint equivariant diffusion over lattices, fractional coordinates and atom types, and **DiffCSP++**¹⁵ adds space-group symmetry constraints, improving physical validity.

5.2 Scaling up: MatterGen, GNoME and language-model generators

The current flagship is **MatterGen**,¹² a diffusion model that generates stable, diverse inorganic crystals across the periodic table by refining atom types, coordinates and lattice together. Decisively for this project, it is *fine-tuned with property labels to steer generation under specified constraints* and outperforms prior models in diversity, novelty and closeness to DFT-derived properties.¹ A complementary, search-driven approach is **GNoME**,⁹ a GNN discovery engine with ensembling and uncertainty quantification that, through an active-learning loop of generation and DFT validation, identified over 2.2 million stable inorganic candidates and tens of thousands of new prototypes, an order-of-magnitude expansion of known stable materials. A third route treats crystals as text: **CrystalLLM**¹⁷ trains a decoder-only Transformer on millions of CIF files and generates new structures by next-token prediction; indeed, even general-purpose LLMs, once fine-tuned, can emit valid, stable inorganic crystals directly as text,⁴² a representation directly compatible with an LLM-centric agent.

RECURRING LIMITATION

Generative models produce *syntactically valid* structures that are not guaranteed to be *synthesizable* or even physically plausible; MatterGen itself cannot guarantee experimental feasibility, and generated structures can be physically implausible despite passing validity checks.^{12,1} This is exactly why the project pairs generation with a property-predictor/verification stage rather than trusting the generator's output directly; synthesizability itself can ultimately only be confirmed experimentally, as autonomous laboratories such as A-Lab³⁴ now begin to do at scale.

6. Datasets, tools and infrastructure

Every model above is only as good as its data, and the project will depend on the same ecosystem. The computational backbone is a set of DFT databases: the **Materials Project**² and the **Open Quantum Materials Database (OQMD)**³, alongside NOMAD, JARVIS, the Open Catalyst sets and QM9, supply hundreds of thousands of structures with computed properties for both training and retrieval.¹ A pivotal caveat, emphasised by Van *et al.*,¹ is **data imbalance**: coverage is heavily skewed toward stable, inorganic, atomically-resolved systems (especially oxides), while polymers, disordered and amorphous phases, soft materials and larger length scales are sparse. This bias shapes what current models can reliably do and bounds the project's claims: a system trained and grounded on Materials-Project-

like data will be most trustworthy for stable inorganic crystals, which appropriately scopes the proposed work.

On infrastructure, mature open-source tools make the project tractable within a year. *pymatgen*³⁶, ASE and the Materials Project / OPTIMADE APIs handle structure manipulation and data access; *matminer* and RDKit provide featurisation; PyTorch Geometric and DGL implement the GNNs; and frameworks such as LangChain and LlamaIndex provide the RAG and agent scaffolding referenced throughout §3.¹ Reference implementations exist for the key models (CGCNN, MEGNet, CDVAE, MatterGen), so the project integrates and adapts proven components rather than rebuilding them, a deliberate de-risking choice reflected in §11.

7. Synthesis: the integration gap and the case for an agent

Three conclusions emerge from the literature. **First**, LLMs are excellent linguistic interfaces but unreliable quantitative reasoners; grounding via RAG fixes *factual* reliability but not *numerical* capability.^{31,7} **Second**, dedicated ML now performs the two quantitative tasks the project needs, property prediction (CGCNN, MEGNet)^{19,20} and property-conditioned crystal generation (CDVAE, ConCDVAE, MatterGen)^{13,16,12}, to a useful standard. **Third**, agent frameworks already show that an LLM can usefully delegate to such tools (LLMatDesign, ChemCrow, ChatMOF),^{26,30,27} but existing systems each fix a narrow toolset and, crucially, *do not unify grounded retrieval, numerical prediction and generative design in one loop*.

THE RESEARCH GAP

Materials RAG systems *retrieve and summarise* but cannot *predict* unmeasured properties or *generate* new structures; property predictors and diffusion generators can do exactly this but are not exposed through a grounded, conversational interface. **No current system lets a researcher ask, in natural language, for a material with a target property and receive a grounded answer that combines retrieved evidence, a predicted property value, and a generated candidate crystal, all verified before it is returned.** Closing this gap is the project.

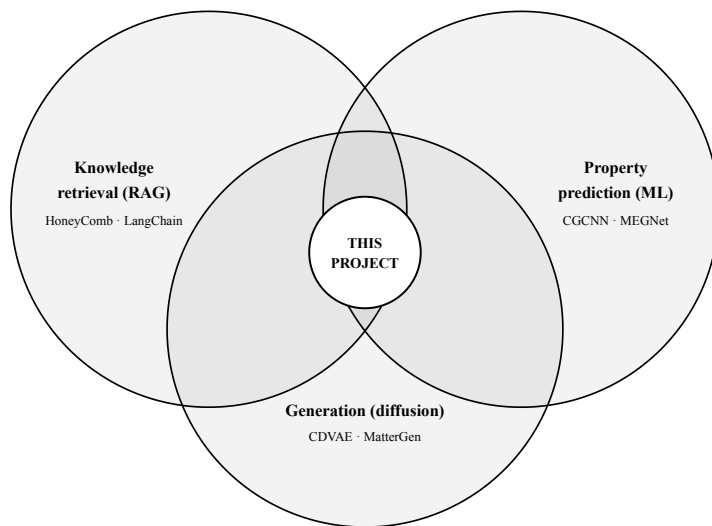


Figure 2. Positioning of the project (original schematic). Each capability is mature in isolation and partially paired by existing agents, but the three-way intersection (grounded retrieval *and* numerical prediction *and* property-conditioned generation in one agentic loop) is largely unoccupied.

An agentic architecture is the natural way to close it because the agent need not *be* good at chemistry; it needs to know *which tool* to call and how to compose the results, exactly the competence LLMs do possess.³⁰ The remainder of this document specifies how.

8. Research proposal: aims and objectives

Background in one paragraph. The project extends an existing materials-RAG framework into a tool-using LLM agent. Where current RAG systems retrieve and summarise materials knowledge, the agent will additionally (i) delegate *property prediction* to a dedicated ML model so it can answer quantitative questions about unmeasured materials, and (ii) delegate *query-based generation* to a diffusion model so it can propose novel crystals that should exhibit a requested property, each result grounded against retrieved data and verified before it is returned.

AIM

To design, build and evaluate an LLM-agent system that unifies retrieval-augmented materials knowledge, ML-based property prediction and diffusion-based crystal generation, enabling natural-language, property-targeted discovery of inorganic crystalline materials with grounded, verifiable outputs.

Objectives.

- O1. Establish the RAG baseline.** Reproduce/adopt a materials-RAG pipeline (knowledge base from Materials Project + literature abstracts; BM25 + dense retrieval) and characterise its limits on quantitative queries.
- O2. Integrate a property predictor as a tool.** Wrap a pretrained GNN (CGCNN/MEGNet) behind the agent so it returns band-gap and formation-energy estimates on demand, benchmarked on Matbench.
- O3. Integrate a generative model as a tool.** Connect a property-conditioned generator (CDVAE/Con-CDVAE, with MatterGen as a stretch target) to produce candidate crystals for a target property.
- O4. Build the agent loop.** Implement routing, tool-calling and a verification stage (predict-then-check, stability filtering) in LangChain/LlamaIndex, with an iterate-if-unmet feedback loop (Fig. 1).
- O5. Evaluate end-to-end.** Define metrics: retrieval faithfulness, prediction error vs DFT (on Matbench³⁷ and LLM4Mat-Bench⁴³), generation validity/novelty/property-match, and hallucination rate, under controlled ablations (LLM-only, retrieval-only, full system) so gains are attributable to each component.

9. Methodology and system design

The system (Fig. 1) is an LLM *orchestrator* exposing three tools through a standard agent framework. **Retrieval (Tool ①):** a knowledge base of Materials Project records and literature abstracts is indexed with a hybrid of sparse (BM25) and dense passage retrieval,⁴⁰ following the HoneyComb pattern;²⁴ the agent cites retrieved passages so every factual claim is traceable. **Prediction (Tool ②):** a pre-trained CGCNN/MEGNet^{19,20} maps a structure (or a retrieved/generated candidate) to a property estimate in milliseconds, supplying the numerical capability the LLM lacks. **Generation (Tool ③):** a property-conditioned diffusion model^{13,16} samples candidate crystals for a requested property; outputs are CIF/structure objects handled with pymatgen.³⁶

The **agent loop** ties these together: the LLM parses the query, decides whether it needs a fact ($\rightarrow$ ①), a number ($\rightarrow$ ②) or a new structure ($\rightarrow$ ③), then ② to score it), composes the results, and passes candidates through a **verification layer** that cross-checks predicted properties against retrieved/known values and applies a stability filter (using a machine-learned potential such as CHGNet⁴¹ to estimate the energy above the convex hull), optionally invoking a single DFT relaxation for the top candidate. If a generated material misses the target property, the agent iterates, re-prompting the generator or proposing a compositional modification in the manner of LLMatDesign.²⁶ Development is incremental and tool-by-tool so that each component is independently testable, and

reuses reference implementations rather than training large models from scratch, feasible within the GPU budget of a single MSci project, in contrast to the tens of thousands of GPU-hours that flagship models such as GNoME and MatterSim required.¹

10. Programme of work

The plan spans the MSci project year (Oct 2026 – Jun 2027), front-loading literature consolidation and the lowest-risk components and deferring the most ambitious generation work, with explicit slack before the dissertation.

Work package	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
WP1 · Literature & tooling set-up									
WP2 · RAG baseline (O1)									
WP3 · Property-predictor tool (O2)									
WP4 · Agent loop + verification (O4)									
WP5 · Generative tool (O3)									
WP6 · End-to-end evaluation (O5)									
WP7 · Dissertation & write-up									

Solid grey bars = core path; black hatching = higher-risk generation work; light hatching = ongoing work / slack. Objectives (O1–O5) map to §8.

11. Risk assessment, fall-backs and flexibility

The plan is deliberately modular so that value is delivered even if the most ambitious objective slips: a working retrieval-plus-prediction agent (O1–O2, O4) is already a complete, assessable outcome, with generation (O3) layered on top.

Table 2. Principal risks and mitigations.

Risk	Severity	Mitigation / fall-back
Diffusion generation too costly / unstable to train or condition	MED	Use a pretrained CDVAE/MatterGen checkpoint rather than training from scratch; fall back to compositional modification + re-prediction (LLMatDesign style) or string-based CrystaLLM generation if conditioning fails.
Generated structures unsynthesisable / physically implausible	MED	Treat generation as hypothesis-only ; gate every candidate through the predictor + stability filter + optional DFT; report property-match and validity rates honestly rather than claiming discovery.
LLM hallucination in the agent's reasoning/answers	HIGH	Mandatory citation of retrieved evidence; numerical answers must come from Tool ② (never the LLM); strict typed function-calling schemas with validation/retry. Fall-back: a deterministic rule-based router maps recognised intents (e.g. "band gap of X") straight to tools, bypassing LLM routing on core paths.
Property predictor inaccurate out-of-distribution	MED	Restrict scope to well-covered inorganic crystals; report predictor uncertainty; cross-validate against retrieved DFT values. Fall-back: the agent abstains (a calibrated "I don't know" using retrieval-coverage and uncertainty signals) rather than answering confidently outside its regime.
Data/compute access (GPU hours, API limits)	LOW	Rely on open data (Materials Project/OQMD) and reference checkpoints; cache retrieved records locally to respect rate limits; College HPC for any DFT. Fall-back: OPTIMADE federation / Alexandria if a single source is unavailable; nothing in the core path needs large-scale pretraining.
Grounding/hallucination metrics noisy or contested	LOW	Use multiple complementary measures and emphasise controlled ablation deltas (LLM-only vs RAG-only vs full), which remain informative even if absolute numbers are imperfect.
Integration/engineering overruns	MED	Tool-by-tool development with independent tests; the RAG + predictor milestone (Jan) is a self-contained deliverable; built-in slack (WP7) before the dissertation.

Flexibility. Decision points are explicit: if generation (WP5) is not yielding valid, property-matched candidates by the end of April, the project pivots to a deeper evaluation of the retrieval-plus-prediction agent (e.g. a broader property set, an ablation of retrieval strategies), which is itself a strong contribution and directly answers the project brief's primary objective.

12. Expected outcomes and deliverables

The project will deliver: (1) a working LLM-agent system integrating RAG, property prediction and (target) generation for inorganic crystals; (2) an evaluation comparing it against bare-LLM and retrieval-only baselines on faithfulness, prediction error, generation quality and hallucination rate; (3) a curated materials knowledge base and a reproducible codebase built on open tools; and (4) the MSci dissertation. The broader contribution is conceptual as much as engineering: a demonstration that the right division of labour (language model for orchestration and explanation, specialised ML for numbers and structures, retrieval for grounding) is a more reliable route to materials discovery than ever-larger general models alone.³¹

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